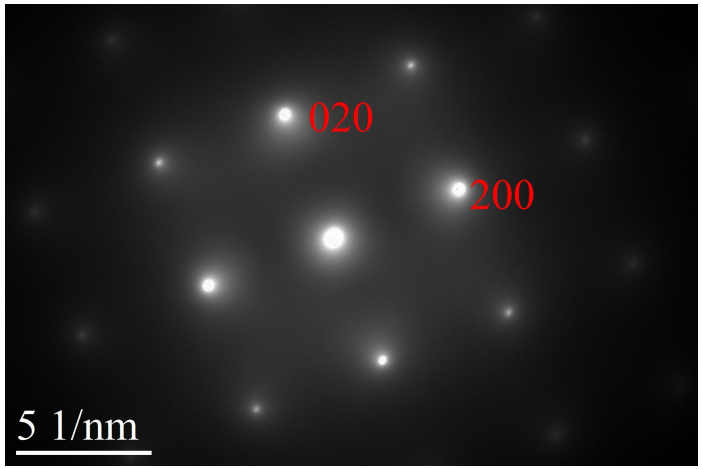
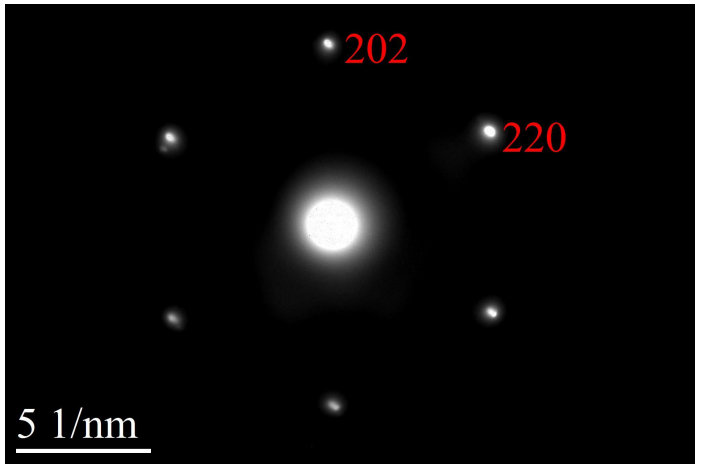
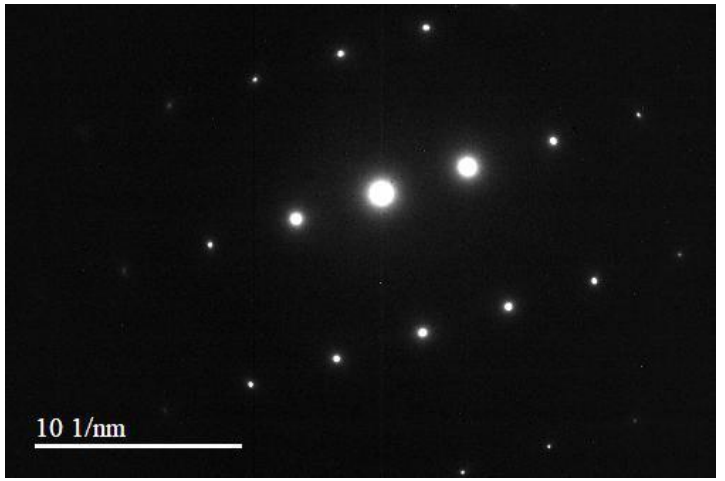
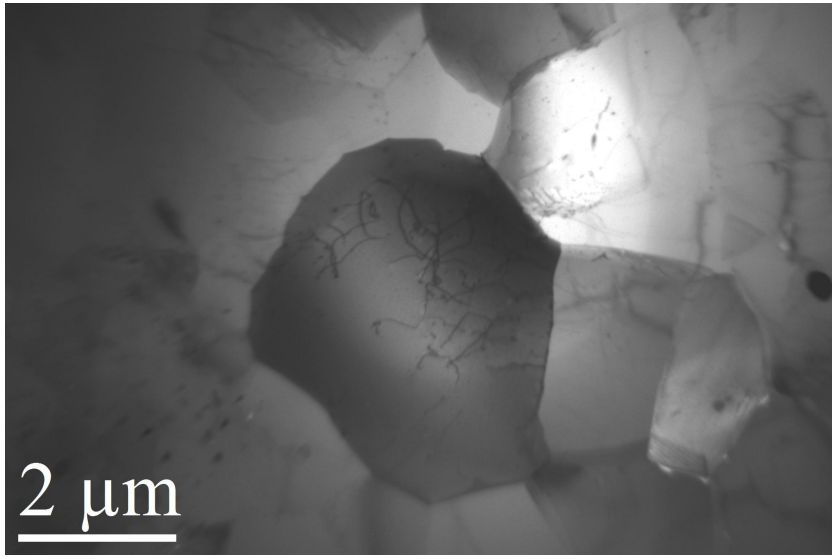
Electron Diffraction Patterns from Al (face center cubic structure)

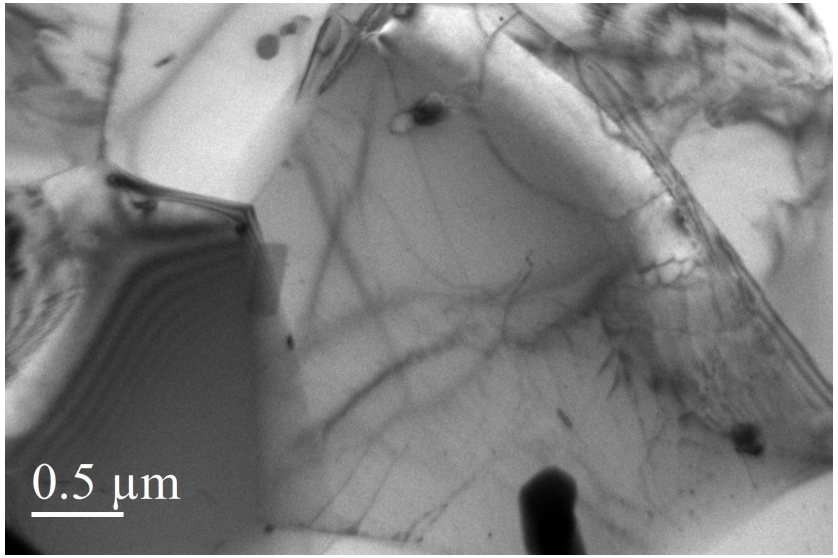
Diffraction patterns along different crystallographic directions of Al

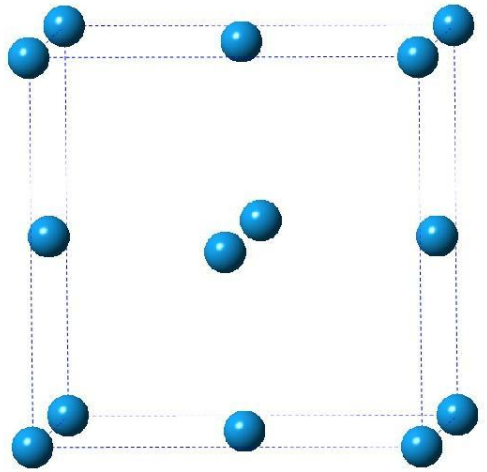












Polycrystalline sample of Al contains grains with different orientations.

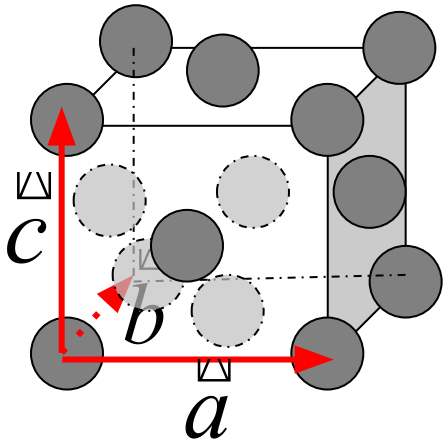
How to analyze diffraction patterns.

A diffraction pattern is a representation of the reciprocal lattice of the material. The spots in the DP correspond to crystallographic directions of the sample and are identified as vectors in reciprocal space called $\vec{g}$ vectors. The intensity in each spot is the result of interference from scattering of the electron wave by the electrostatic potential of each atom in the crystal.

Measure distances between spots
Measure angles between $\vec{g}$ vectors
Compare distances and angles to theoretical values from ICSD data set for the material, if known.

Real and Reciprocal Space

Real Space lattice

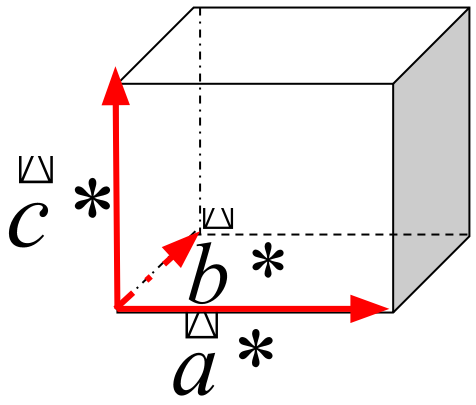


$$\vec{a} = a\,\hat{x}$$
$$\vec{b} = a\,\hat{y}$$
$$\vec{c} = a\,\hat{z}$$

Basis:
Al at (000) (1/2,1/2,0)
(1/2,0,1/2) (0,1/2,1/2)

Units are in nm

Reciprocal lattice



$$\vec{a}^* = \frac{\vec{b}\times\vec{c}}{\vec{a}\bullet(\vec{b}\times\vec{c})} = \frac{1}{a}\hat{x}$$

$$\vec{b}^* = \frac{\vec{c}\times\vec{a}}{\vec{a}\bullet(\vec{b}\times\vec{c})} = \frac{1}{b}\hat{y}$$

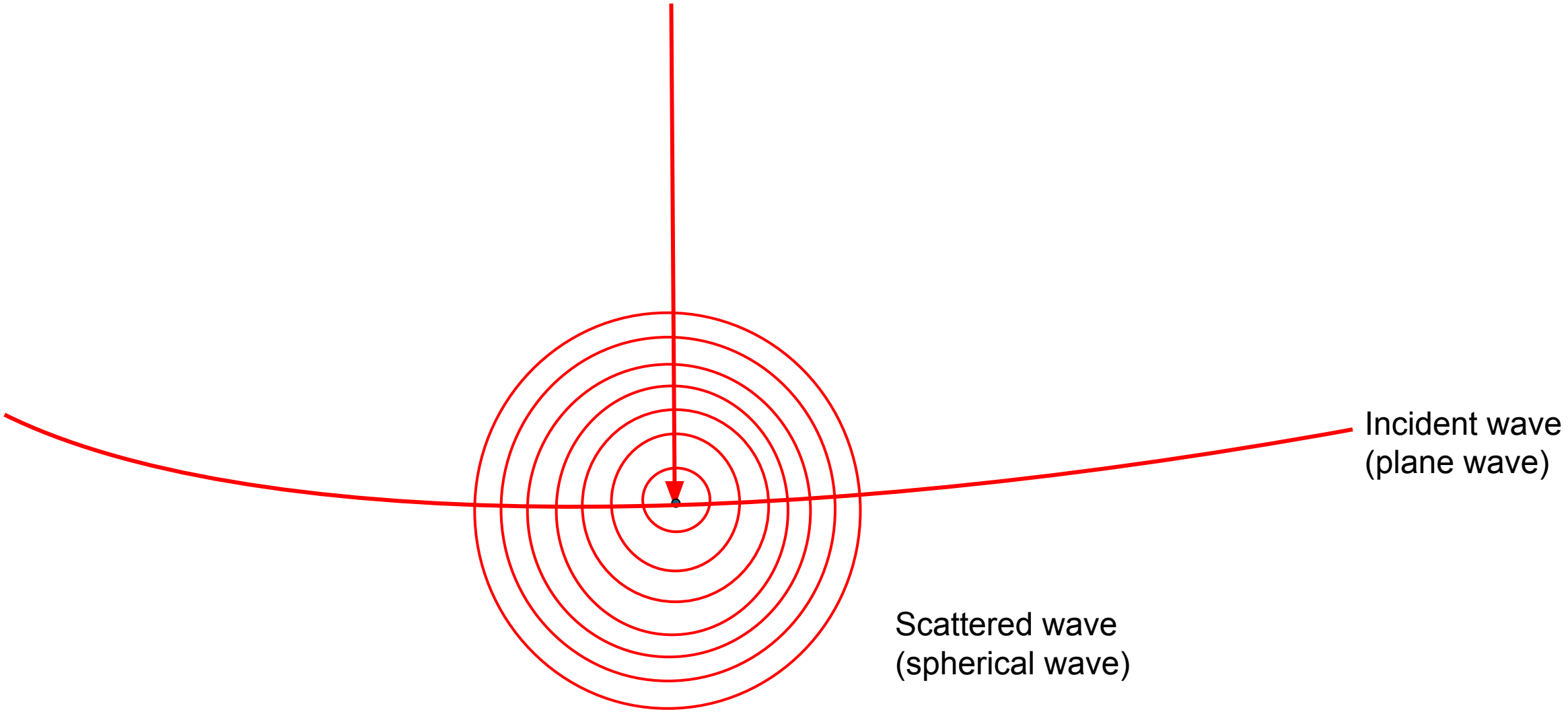
$$\vec{c}^* = \frac{\vec{a}\times\vec{b}}{\vec{a}\bullet(\vec{b}\times\vec{c})} = \frac{1}{c}\hat{z}$$

There is no basis in reciprocal space

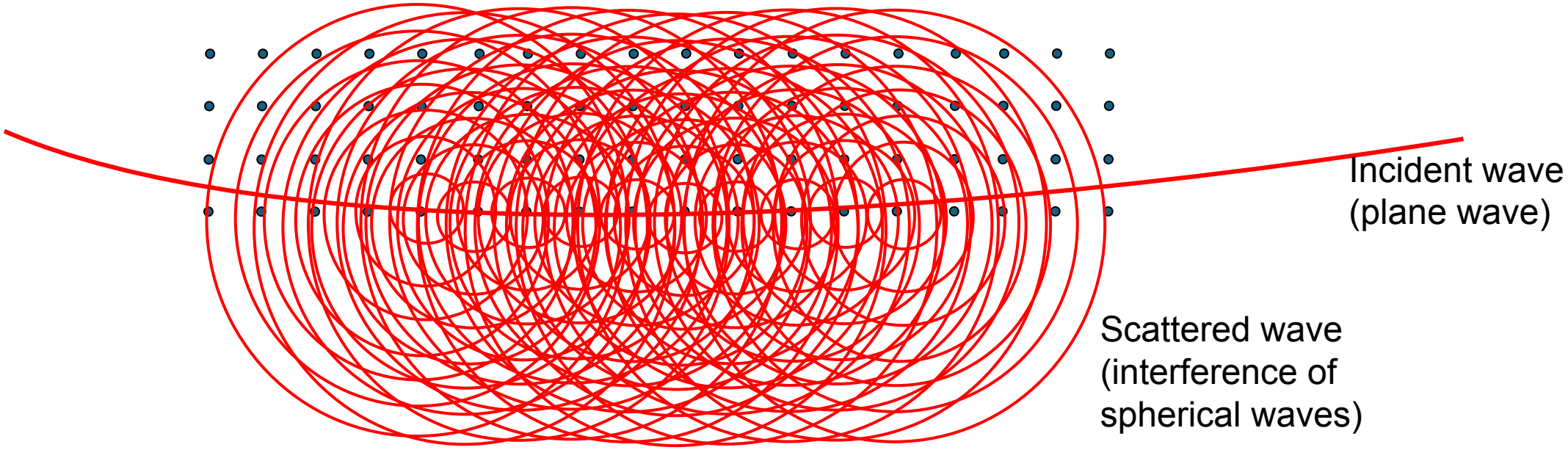
Units are in nm^{-1}

Both real and reciprocal space lattices are periodic

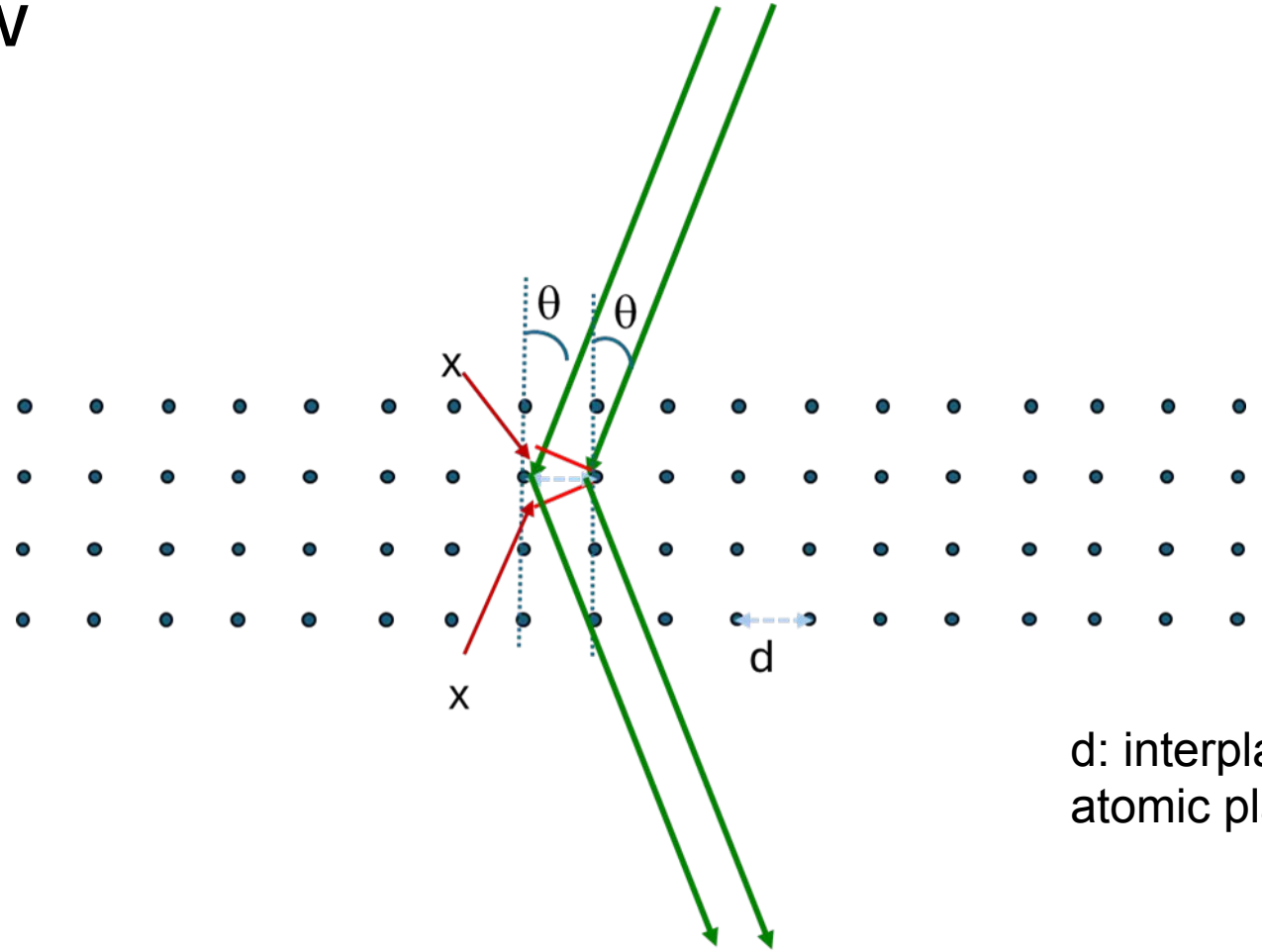
Scattering of electrons by the atomic potential of an atom $V(\vec{r})$



Scattering of electrons by the potential of a crystal



Bragg's Law



d: interplanar distance between atomic planes

Path difference between two beams $2x=2d\sin(\theta)=n\lambda$ for constructive interference

For electrons $\lambda = 0.0025$ nm, d ~ 0.2 nm, $\theta < 1°$

$$2\theta \approx n\lambda\frac{1}{d}$$

Bragg's Law in reciprocal space

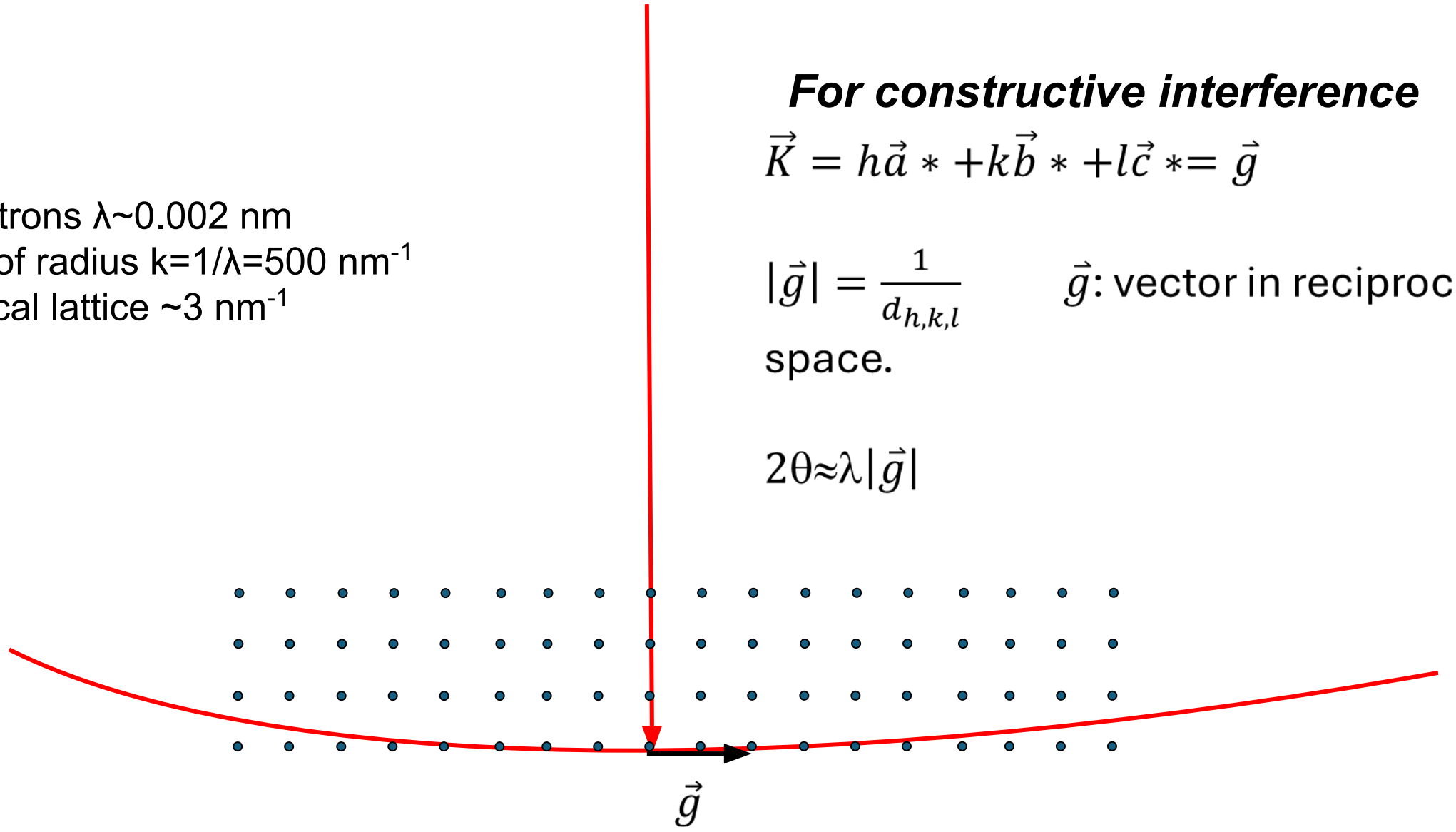
For electrons λ~0.002 nm
Sphere of radius k=1/λ=500 nm^{-1}
Reciprocal lattice ~3 nm^{-1}

For constructive interference

$$\vec{K} = h\vec{a}* + k\vec{b}* + l\vec{c}* = \vec{g}$$

$|\vec{g}| = \frac{1}{d_{h,k,l}}$ $\vec{g}$: vector in reciprocal space.

$2\theta \approx \lambda|\vec{g}|$



Diffraction from Quantum Mechanics

$$\phi(\vec{g}) = \frac{i\lambda F_g}{V_c \cos\theta}\left(\int e^{-2\pi i \vec{g}\cdot\overrightarrow{r_n}}\right)$$

$$F_g = \sum_{i\ in\ u.c.} f_i e^{2\pi i \vec{g}\cdot\vec{r}_i}$$

$$f_i = \frac{m_e}{2\pi\hbar^2}\int V_a(\vec{r}) e^{2\pi i \vec{K}\cdot\vec{r}} d^3r$$

λ: wavelength of electron beam

F_g: structure factor

V_c: volume of unit cell

θ: Bragg angle

$\vec{r}_n = n_1\vec{a} + n_2\vec{b} + n_3\vec{c}$: positions of unit cells in crystal

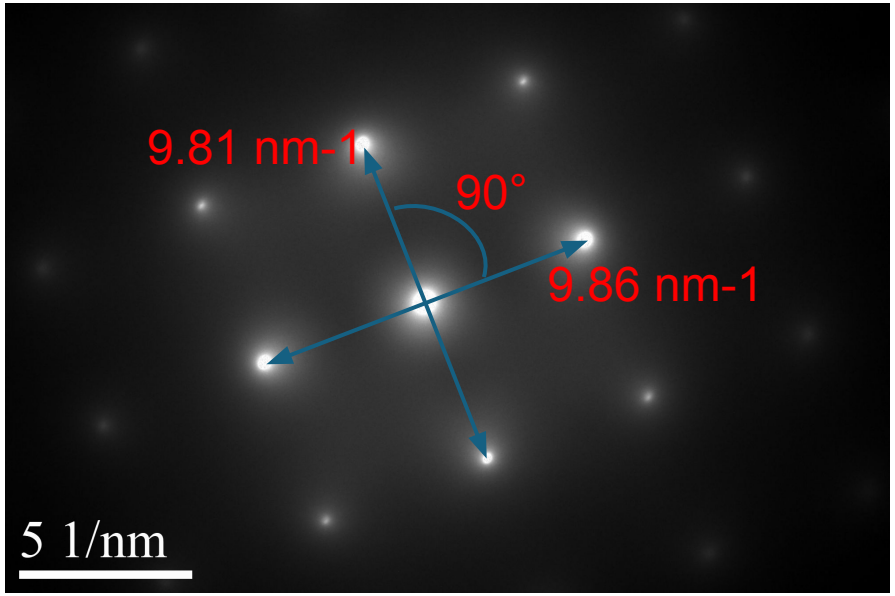
f_i: atomic form factor for atom i

$\vec{r}_i$: atom positions in unit cell

V_a: atomic potential for atom I

$\vec{K}$: vector in reciprocal space

Examples



See ICSD file for Al with symmetry operators to find what reciprocal lattice spots have non-zero intensity due to the symmetry of the crystal structure of Al.

Take inverse of distances in DP divided by 2

$$\frac{1}{9.81/2} = 0.204\ nm$$

$$\frac{1}{9.86/2} = 0.203\ nm$$

Published Crystal Structure Data

Cell parameter	4.0494 4.0494 4.0494 90. 90. 90.	Space group	F m -3 m (225)
Cell volume	66.40 Å³	Z	4
Crystal system	cubic	Crystal class	m-3m
Laue class	m-3m		
Structure type	Cu		
Pearson symbol	cF4		
Wyckoff sequence	a		
Calc. density	2.70 [g/cm³]		

	a/b	b/c	c/a
Axis ratios	1.0000	1.0000	1.0000

EL	Lbl	OxState	Wyck Symb	X	Y	Z	SOF	B
Al	1	+0.00	4 a	0	0	0	1.000000	0.8

Indexing Diffraction Patterns

See ICSD file for Al with symmetry operators to find what reciprocal lattice spots have non-zero intensity due to the symmetry of the crystal structure of Al.

1. Take inverse of distances in DP divided by 2

$$\frac{1}{9.81/2} = 0.204\ nm$$

$$\frac{1}{9.86/2} = 0.203\ nm$$

2. Measure angles between vectors in DP

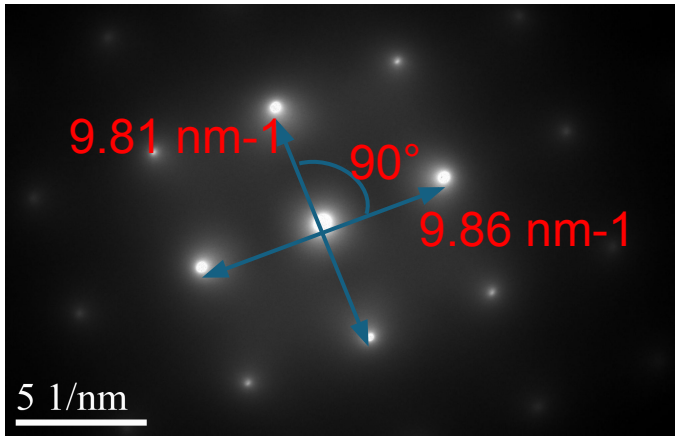
90° in this example

2. Compare distances to theoretical distances for Al.

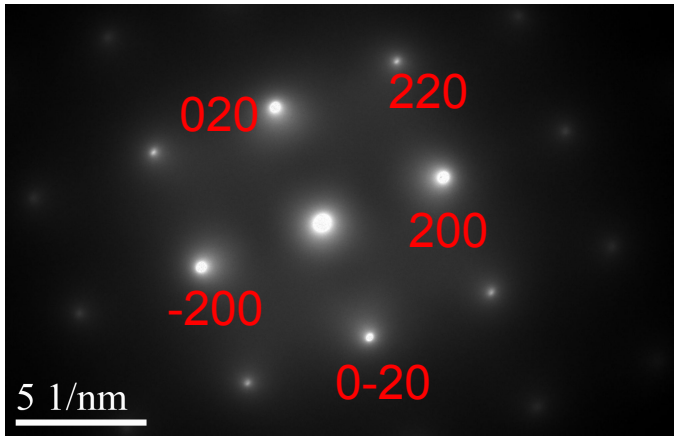
$$d_{h,k,l} = \left\{\frac{h^2+k^2+l^2}{a}\right\}^{-1}$$ where a =lattice constant of Al

These distances can be found in tables with their relative intensities for Al

Analysis for Al DP

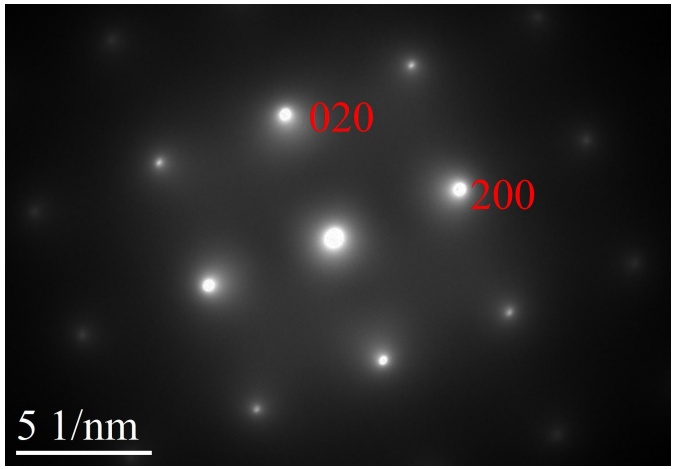


Dist in DP (nm^{-1})	Exp.n Dist. Real space (nm)	Possible hkl	select
9.81	0.204	200, 020, 002	200
9.86	0.203	200, 020, 002	020
13.91	0.144	220, 022, etc	220

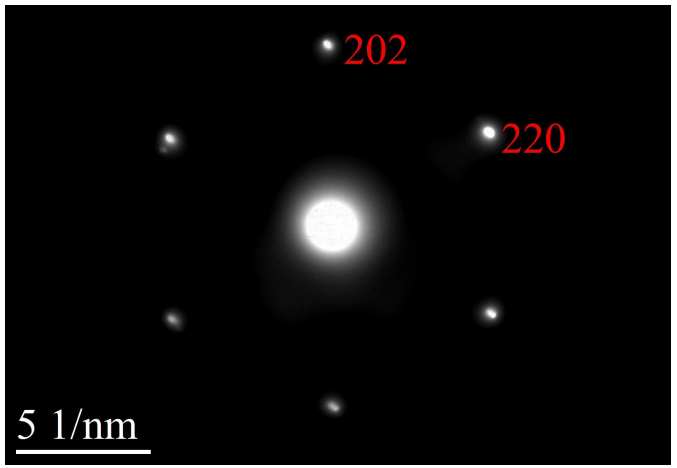


hkl	d_{hkl} (nm)	F_{hkl}
	0.4049	0
	0.2025	$=4f_{Al}$
	0.286	0
	0.143	$=4f_{Al}$

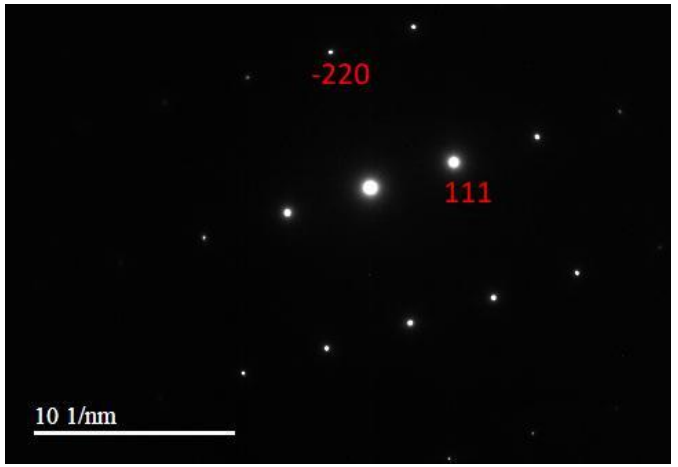
Each DP corresponds to a different orientation of the crystal with respect to the e-beam



[001] projection



[-111] projection



[11-2] projection

